

# Lab 2: Decision Tree Modeling and Improvement

## Predicting Student Dropout and Academic Success

Group [TODO — fill in GroupID]  
[TODO — Course name / instructor, if required by the submission template]

August 30, 2026

### 1 a. Group Introduction

[TODO — Group name and GroupID]

Table 1: Team members and contributions

| Name            | Student ID  | Contribution                                                                                                                                                   | %          |
|-----------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| [TODO — Name A] | [TODO — ID] | Data collection & description, EDA, data preprocessing pipeline, final results aggregation, wrote Introduction / Dataset Description / Comparison / Conclusion | [TODO — %] |
| [TODO — Name B] | [TODO — ID] | Baseline model, tree/rule visualization tooling, wrote Baseline Model & Analysis of the Tree                                                                   | [TODO — %] |
| [TODO — Name C] | [TODO — ID] | Cost-complexity pruning (Improvement 1), wrote section f.1                                                                                                     | [TODO — %] |
| [TODO — Name D] | [TODO — ID] | Class imbalance handling (Improvement 2), wrote section f.2                                                                                                    | [TODO — %] |
| [TODO — Name E] | [TODO — ID] | Feature selection for early warning (Improvement 3), slides, video, references, wrote section f.3                                                              | [TODO — %] |

[TODO — Adjust the contribution text/percentages to match what each person actually did – cross-check against git log and progress/\*.md before finalizing, not just this template.]

### 2 b. Introduction

#### 2.1 Brief introduction to decision trees

A decision tree is a supervised learning model used for both classification and regression tasks. It represents the decision process as a tree structure: each *internal node* tests one feature (e.g. “Tuition fees up to date  $\leq 0.5$ ?”), each *branch* corresponds to the outcome of that test, and each *leaf node* assigns a predicted class.

Training a decision tree follows a **greedy, top-down, recursive-partitioning** strategy: at each node, the algorithm evaluates every feature and every possible split threshold, and picks the split that reduces class *impurity* the most, then repeats recursively on each resulting subset. Two impurity measures are standard:

- **Entropy / Information Gain** – used by the ID3 algorithm (Quinlan, 1986), measuring the information gained by a split.
- **Gini impurity** – used by CART (Breiman et al., 1984), measuring the probability of misclassification if a label were drawn at random from the node’s class distribution.

The scikit-learn library used in this project implements an optimized variant of CART (Pedregosa et al., 2011); it defaults to Gini but supports switching to entropy via the `criterion` parameter.

The **main strength** of a decision tree compared to “black-box” models (neural networks, complex ensembles) is **interpretability**: every path from the root to a leaf can be read directly as an IF (condition 1) AND (condition 2) ... THEN (label) rule, with no separate explanation technique (SHAP, LIME) required. Trees also handle both numerical and categorical features natively and require no feature scaling – both properties are exploited directly in Section 3 below.

The **main weakness** is a tendency to **overfit**: an unconstrained tree keeps splitting until almost every leaf is pure on the training set, effectively memorizing training noise, which yields very high training accuracy but a much lower test accuracy (Mitchell, 1997; Breiman et al., 1984). This is exactly what is observed for the baseline model in Section 4, and is the motivation for the pruning improvement in Section 6.

## 2.2 Objective of the project

The project applies this theory to a real, socially meaningful problem: **predicting whether a student drops out, remains enrolled longer than expected, or graduates on time**, using real data from 4,424 university students in Portugal. Concretely, the group:

1. Builds and evaluates an unconstrained baseline decision tree, measuring performance with the full set of metrics appropriate for a 3-class classification problem (Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix).
2. Analyzes the baseline tree’s structure: what the root split is and why the algorithm chose it, whether the tree shows signs of overfitting, and extracts concrete decision rules.
3. Proposes and tests three improvement methods, each targeting a different weakness of the baseline:
  - **Pruning** (cost-complexity pruning) – addresses overfitting, trading tree complexity for generalization.
  - **Class-imbalance handling** (`class_weight='balanced'`, SMOTE) – addresses the model’s default bias toward the majority class (Graduate), which causes it to miss the minority classes that matter most for the application (Enrolled, Dropout).
  - **Feature selection for early warning** – addresses a practical limitation: the baseline’s strongest features are only known *after* a semester ends, so the most accurate model is not necessarily the model that is actually deployable for early intervention.

4. Quantitatively compares all four models and draws conclusions not only about the numbers, but about **which model suits which application goal** – in the spirit the assignment brief emphasizes: not just running a model, but explaining how the tree is built, how it behaves, and why each improvement helps (or does not).

## 3 c. Dataset Description

### 3.1 Data source

The group uses the “**Predict Students’ Dropout and Academic Success**” dataset from the UCI Machine Learning Repository (ID = 697), published by Realinho, Vieira Martins, Machado, and Baptista (Realinho et al., 2021), licensed CC BY 4.0. The dataset is described in detail in the accompanying paper (Realinho et al., 2022).

The data was collected at **Instituto Politécnico de Portalegre**, Portugal, spanning **17 undergraduate degree programs** (Agronomy, Design, Nursing, Journalism and Communication, Management, Social Service, Biofuel Production Technologies, Informatics Engineering, etc.), academic years **2008/09 to 2018/19**. It combines information from several administrative sources at enrollment time (admission records, socio-economic data) with academic outcomes at the end of each of the first two semesters.

### 3.2 Dataset description — numbers confirmed from actual data

The group did not copy figures from the UCI metadata page; instead it directly inspected the raw dataset to confirm:

- The raw file contains **4,424 rows and 37 columns** → 4,424 samples (students), 36 features + 1 label column **Target**.
- There are **zero** missing values across the whole dataset (the authors pre-cleaned the data).
- **Notable finding:** the UCI page lists 36 features, but Table 1 of the original paper lists only 34. Cross-checking the actual column list against that table, the discrepancy is exactly two columns: **Previous qualification (grade)** and **Admission grade** — present in the released CSV but absent from the paper’s table. This is direct evidence the group inspected the real data rather than copying the metadata.

The 36 features fall into six groups by nature: demographics (6 columns, e.g. Gender, Age at enrollment), socio-economic status (8 columns, e.g. Tuition fees up to date, Scholarship holder), macroeconomics (3 columns: Unemployment rate, Inflation rate, GDP), academic background at enrollment (7 columns, e.g. Course, Admission grade), first-semester outcomes (6 columns), and second-semester outcomes (6 columns).

**Target variable** has 3 classes, moderately imbalanced:

Table 2: Target class distribution ( $n = 4,424$ )

| Class    | Count | %     | Meaning                                                     |
|----------|-------|-------|-------------------------------------------------------------|
| Graduate | 2,209 | 49.9% | Graduated on the standard timeline (3 years; 4 for Nursing) |
| Dropout  | 1,421 | 32.1% | Dropped out                                                 |
| Enrolled | 794   | 17.9% | Still enrolled past the standard timeline                   |

**Enrolled** is the minority class and the hardest to classify, as it sits “between” the other two classes in feature space – this is the reason the class-imbalance improvement (Section 6, f.2) exists.

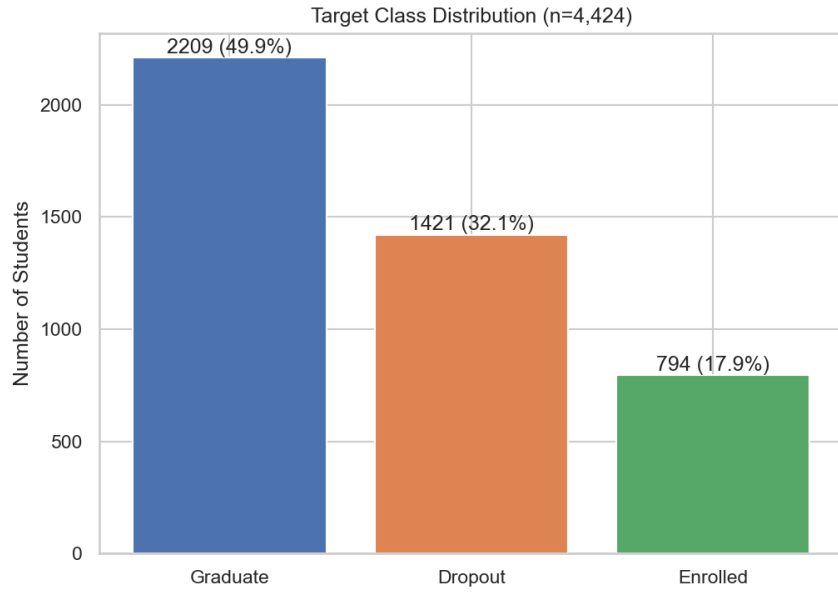


Figure 1: Target class distribution ( $n = 4,424$ ).

Several of the observations below refer to how strongly a feature drives the baseline decision tree’s actual splits, measured by Gini importance (the total reduction in class impurity attributed to that feature across the whole tree, described further in Section 4). Table 3 lists the top 10 features by this measure, plus one lower-ranked feature included for contrast with the discussion below.

Table 3: Top-10 features by Gini importance in the baseline tree (M0), plus one feature shown for contrast

| Rank | Feature                                | Gini importance |
|------|----------------------------------------|-----------------|
| 1    | Curricular units 2nd sem (approved)    | 0.340           |
| 2    | Curricular units 2nd sem (grade)       | 0.045           |
| 3    | Tuition fees up to date                | 0.045           |
| 4    | Curricular units 2nd sem (enrolled)    | 0.045           |
| 5    | Admission grade                        | 0.044           |
| 6    | Course                                 | 0.044           |
| 7    | Curricular units 1st sem (evaluations) | 0.034           |
| 8    | Previous qualification (grade)         | 0.031           |
| 9    | Curricular units 1st sem (approved)    | 0.029           |
| 10   | Mother’s occupation                    | 0.028           |
| ...  |                                        |                 |
| 24   | Scholarship holder                     | 0.010           |

Descriptive statistics from exploratory data analysis illustrate how strongly some features relate to the target: among students who had **not paid tuition on time**, 86% ended up in the Dropout class, versus 25% Dropout / 56% Graduate among students who had paid on time. Students with a **scholarship** have a 3× lower Dropout rate than those without (12% vs. 39%).

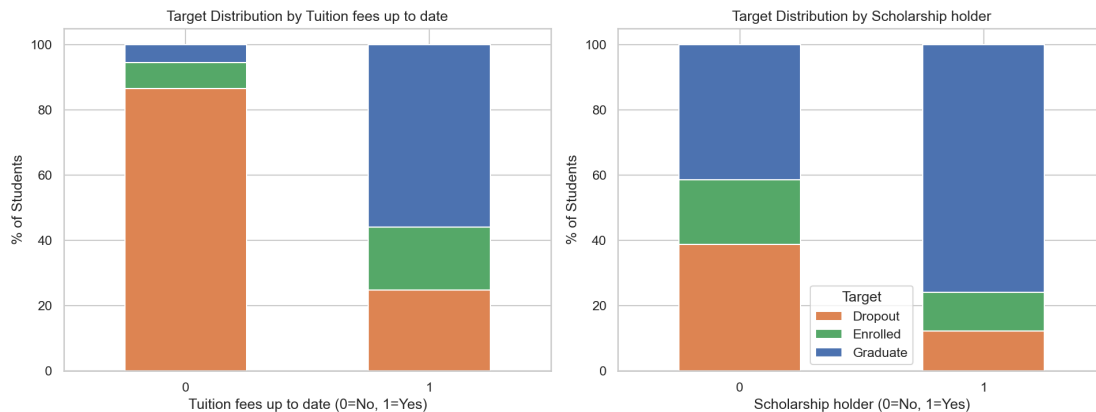


Figure 2: Target distribution split by Tuition fees up to date (left) and Scholarship holder (right).

Tuition fees up to date is among the top-5 features by permutation importance in the original paper (Realinho et al., 2022), and is also rank 3 in Table 3 – the two ways of measuring importance agree here. Scholarship holder is a more subtle case: it is **not** in the top-5 of either the paper or Table 3 (rank 24), even though its percentage-of-target chart above looks just as striking as tuition’s. A large gap in a simple percentage-of-target chart does not automatically mean a feature is an important tree split once it is placed alongside stronger, possibly overlapping features such as Tuition fees up to date – the tree may be extracting the same signal from a stronger correlate instead.

By program of study, the two lowest-graduation programs are course code 33 (Biofuel Production Technologies, 66.7% Dropout) and course code 9119 (Informatics Engineering, 54.1% Dropout) –

only **8.3%** and **8.2%** of their students graduate respectively, versus the dataset average of 49.9%. The two highest-graduation programs are course code 9238 (Social Service, 69.9% Graduate) and course code 9500 (Nursing, 71.5% Graduate), with the lowest Dropout rates (18.3% and 15.4%).

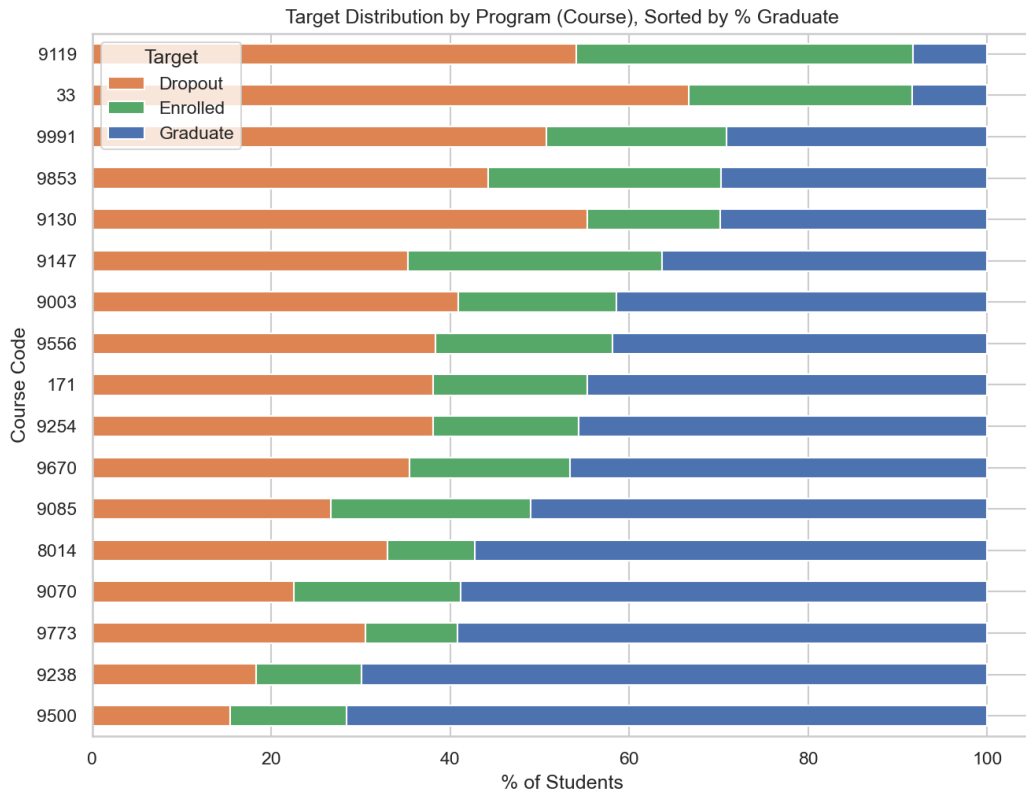


Figure 3: 17 programs sorted by increasing Graduate rate.

The original paper ranks **Course** in its top-5 features by permutation importance ([Realinho et al., 2022](#)); in Table 3, **Course** ranks 6th – close to top-5 either way, and importantly, available at enrollment time, which is why it underpins the early-warning model M3 (Section 6, f.3).

The strongest single predictor in the whole dataset, by contrast, is only known once a semester has actually happened:

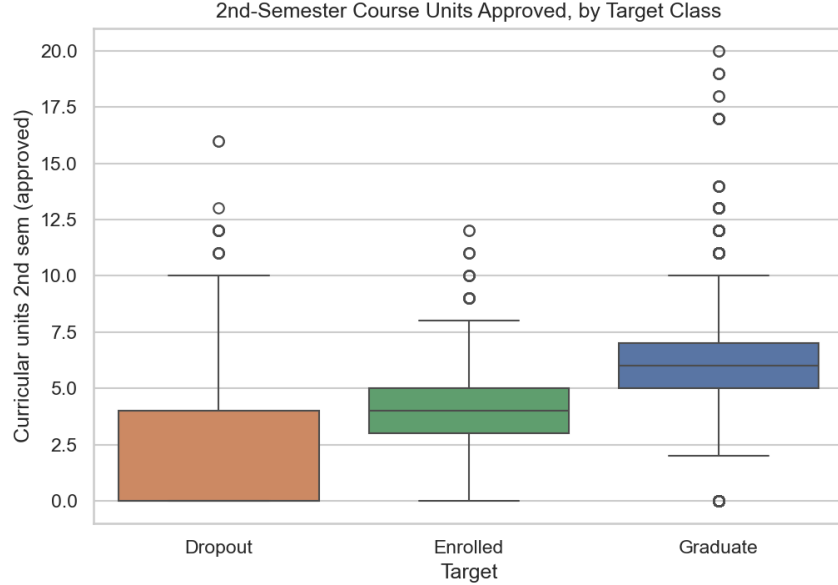


Figure 4: Curricular units 2nd sem (approved) by Target class.

Median values are 0 for Dropout, 4 for Enrolled, and 6 for Graduate. This is the single most important feature in Table 3 (rank 1, importance 0.340 – more than the sum of ranks 2–5 combined), but it is only known *after* the second semester ends, so it cannot be used by the early-warning model M3.

### 3.3 Preprocessing performed

All preprocessing logic is implemented in a single shared pipeline, used by the whole team so every model is evaluated on the exact same train/test split.

1. **No missing-value handling needed** – the dataset is already clean at the source (0 missing values).
2. **Re-encoding categorical columns stored as integers.** Many columns are integer codes for unordered categories (e.g. **Course** code 33 = Biofuel Production Technologies, code 9500 = Nursing – these codes carry no ordinal relationship). Left as-is, scikit-learn would treat them as continuous and produce meaningless splits like **Course**  $\leq 8.5$ . The group used a mixed strategy:
  - **One-hot encode** the 4 low-cardinality columns: **Marital Status** (6), **Application mode** (18), **Course** (17), **Previous qualification** (17) – so each split reads as a clear yes/no question (e.g. “is this the Nursing program?”).
  - **Keep the raw integer code** for high-cardinality columns: **Mother’s/Father’s occupation** (32/46 values), **Mother’s/Father’s qualification** (29/34), **Nacionality** (21) – one-hot encoding these would create dozens of sparse columns, destabilizing the tree more than it would help. This trade-off is noted explicitly as a limitation.
  - After this step the feature count grows from 36 to **90** columns.

3. **No feature scaling.** A decision tree splits on a threshold of one feature at a time, not on distances between data points as KNN or linear regression do, so it is invariant to any monotonic per-column transform such as scaling. Adding `StandardScaler/MinMaxScaler` here would be redundant and would not change the learned tree structure at all.
4. **Stratified train/test split:** 80%/20%, stratified by class to preserve the  $\approx 50/32/18$  class ratio in both sets – required because the data is imbalanced, to avoid a random split accidentally starving the test set of the minority class. A fixed random seed is used so all 5 models in the project are evaluated on the exact same split.

After preprocessing: **3,539 training samples / 885 test samples**, each with 90 numeric features, ready to feed directly into a decision tree classifier.

### 3.4 Multicollinearity among features

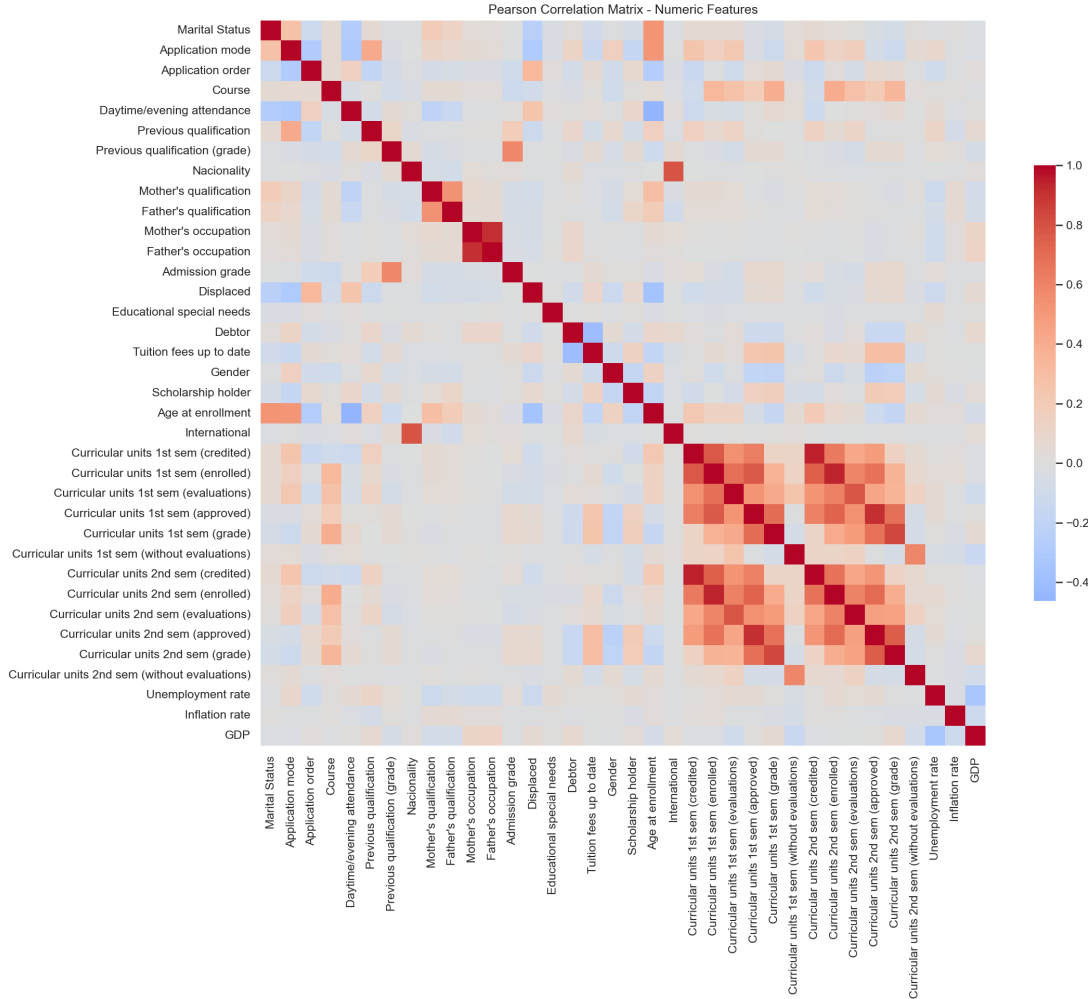


Figure 5: Pearson correlation heatmap across all numeric features.

The heatmap's colors show the overall pattern, but individual cell values are not printed on a  $36 \times 36$  grid; Table 4 lists the exact values for the pairs discussed here so they can be checked directly.

Table 4: Highest Pearson correlations in the dataset

| Feature                             | Correlated with                     | $r$   |
|-------------------------------------|-------------------------------------|-------|
| Curricular units 1st sem (credited) | Curricular units 2nd sem (credited) | 0.945 |
| Curricular units 1st sem (enrolled) | Curricular units 2nd sem (enrolled) | 0.943 |
| Mother’s occupation                 | Father’s occupation                 | 0.910 |
| Curricular units 1st sem (approved) | Curricular units 2nd sem (approved) | 0.904 |
| Nacionality                         | International                       | 0.791 |

The three highest-correlation pairs in the whole dataset are `1st sem credited`  $\leftrightarrow$  `2nd sem credited`, `1st sem enrolled`  $\leftrightarrow$  `2nd sem enrolled`, and `Mother’s occupation`  $\leftrightarrow$  `Father’s occupation` – this last pair is *higher* than the semester-approved pair (`1st/2nd sem approved`). The pair `Nacionality`  $\leftrightarrow$  `International` is still a high correlation (international students almost always have a non-Portuguese `Nacionality` code), but lower than the four pairs above.

**[TODO — Data-integrity note (delete once acted on): the multicollinearity table in docs/02-DATASET-VA-CONG-VIEC.md Section A6 has two values that do not match the real data when recomputed: it lists `Nacionality` $\leftrightarrow$ `International` as  $r = 0.912$  (actual  $r = 0.791$ ), and `Mother’s/Father’s occupation` as  $r = 0.724$  (actual  $r = 0.910$  – *much higher* than recorded, in fact the 3rd-highest pair in the dataset, not rank 9 as the old table implies). This report uses the recomputed values above; docs/02 should be updated so Role E’s rationale for dropping columns in f.3 is not built on the wrong ranking.]**

A decision tree is not as sensitive to multicollinearity as linear regression (it does not need redundant features removed to converge), but multicollinearity still splits feature importance across correlated columns and can make the tree structure less stable across runs. This is a quantitative basis for Role E to consider dropping a redundant column (e.g. `International`, or one of the two parent-occupation columns) when extending the feature-selection experiment in Section 6 (f.3).

### 3.5 Why this dataset is appropriate for decision tree modeling

1. **Features have clear, real-world semantics** (tuition, scholarship, semester outcomes, program of study, ...), so every split translates into an ordinary sentence – this plays directly to a decision tree’s interpretability strength, unlike image or text data where that advantage is less exploitable.
2. **A mix of categorical and numerical features** – a tree handles both natively without scaling, unlike KNN or linear models which require it.
3. **A genuine class imbalance** (50%/32%/18%) – creates a realistic scenario to apply and compare imbalance-handling techniques, one of the improvement directions the brief itself suggests (Section 3.2.d, “Handling class imbalance”).
4. **Features split cleanly by collection time** (known at enrollment vs. known only at end of semester) – enables a feature-selection experiment with real application meaning (early warning), not just a statistics-driven selection.

5. **No missing values** – lets the group spend its time on analysis and interpretation instead of data cleaning, matching the brief’s own emphasis: not just running a ready-made model, but clearly explaining how the tree is built and why the improvements help.

### 3.6 Ethical note

The dataset contains sensitive variables: gender, nationality, marital status, and parents’ education/occupation. Any tree splits on these variables only reflect **correlations observed in one Portuguese university’s sample from 2008–2019**, not causal relationships, and should not be used as grounds for any discriminatory policy toward students in practice.

## 4 d. Baseline Model

### 4.1 Model description and training procedure

[**TODO — Describe the baseline model** (`DecisionTreeClassifier(random_state=42)`, no constraints, Gini criterion), and the train/test procedure using the shared `src.data.get_train_test()` split (3,539 train / 885 test).]

### 4.2 Resulting tree

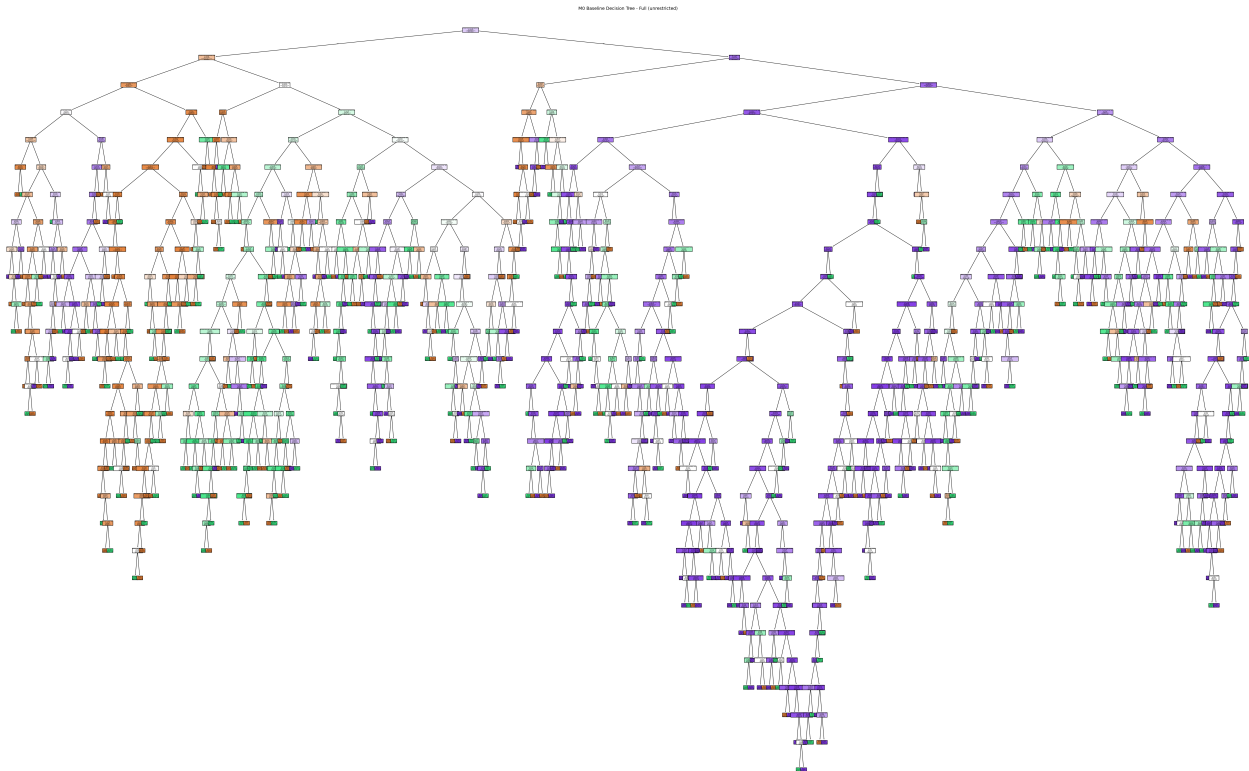


Figure 6: [**TODO — Caption: full M0 tree – state node/leaf count and what it illustrates (overfitting).**]

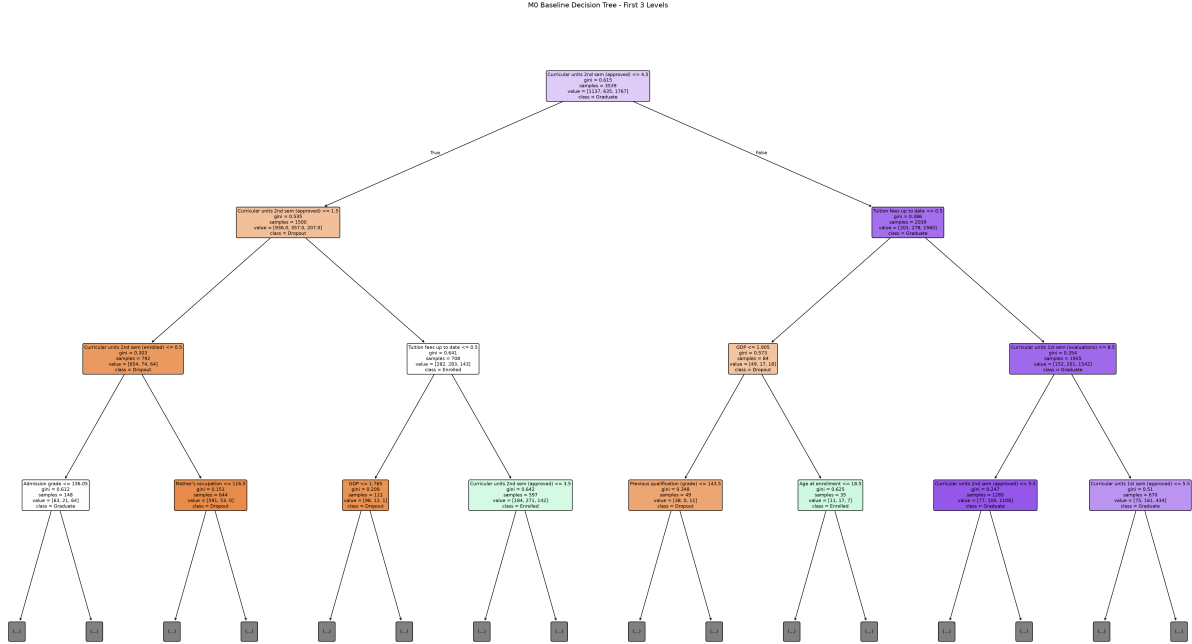


Figure 7: [TODO — Caption: M0 tree truncated to max\_depth=3 for readability – describe the root and first splits shown.]

### 4.3 Accuracy and error rate

[TODO — Report train\_acc, test\_acc, error\_rate from outputs/results.csv (M0 row) and outputs/classification\_report\_M0.txt. Include the confusion matrix figure below.]

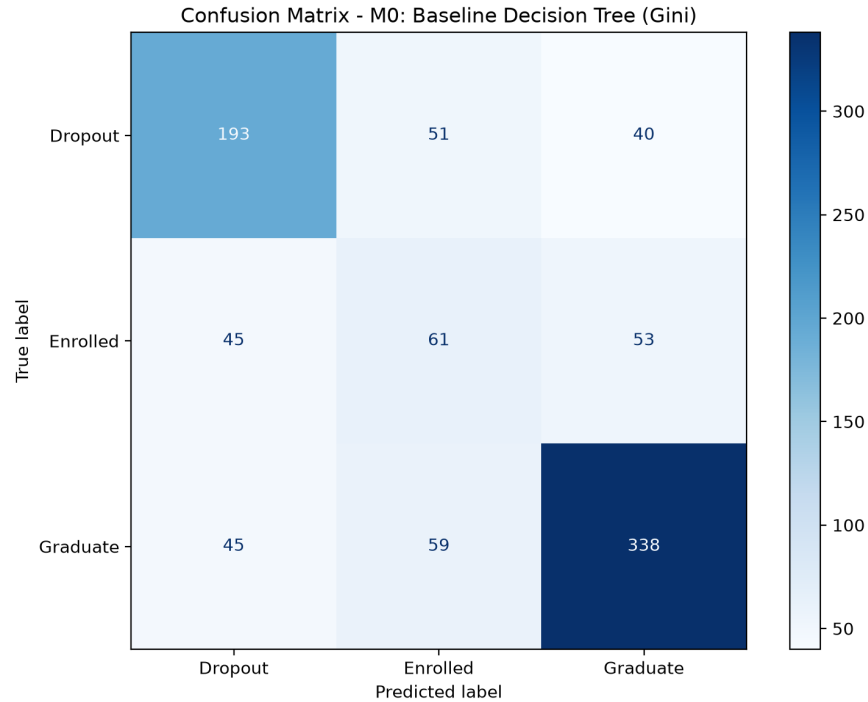


Figure 8: [TODO — Caption: confusion matrix on the 885-sample held-out test set.]

[TODO — Optional: the Gini-vs-Entropy quick check mentioned in docs/02-DATASET-VA-CONG-VIEC.md C0 note – one sentence is enough, this is explicitly not one of the 3 official improvements.]

## 5 e. Analysis of the Tree

### 5.1 Root split

[TODO — Which feature/threshold is the root split, its Gini before/after, and why the algorithm picked it over alternatives.]

### 5.2 Root and the next three levels

[TODO — Table or description of the first 3-4 levels: split feature, threshold, sample counts, class distribution, predicted class at each node – translate the verified node table from the source draft.]

### 5.3 Tree complexity

[TODO — Total node/leaf count, % of leaves with 1-2 samples, average leaf depth – and what that says about overfitting risk.]

### 5.4 Overfitting evidence

[TODO — Train vs. test accuracy gap (generalization gap), and whether it falls in the expected 0.65–0.72 test-accuracy band from AGENT.md (a red flag if not).]

## 5.5 Representative IF-THEN rules

[TODO — Translate the 3 rules from the source draft verbatim (Dropout, Graduate, Enrolled), each with the full condition chain, leaf sample count  $n$ , and purity. Do not paraphrase the conditions loosely – copy the exact feature/threshold values, they were independently re-verified against outputs/rules\_M0.txt.]

*Example rule format (replace with the real translated rules):*

IF Curricular units 2nd sem (approved)  $\leq 1.5$  AND ...  
THEN Dropout ( $n = 189$ , purity 100%).

## 5.6 Strengths and weaknesses of the baseline tree

[TODO — Summarize: strengths (readable top splits, handles nonlinearity, no scaling needed) vs. weaknesses (hard to present as a whole, unstable predictions, especially weak on Enrolled, uses semester-outcome features so not usable for early warning).]

[TODO — Ethical/limitations note (see source draft’s own closing section): sensitive features (gender, Nationality, marital status, parents’ occupation/education) appear in some rule conditions – these must be flagged as correlational, not causal, and not used as grounds for discrimination.]

# 6 f. Improvement Methods

## 6.1 f.1 Improvement Method 1: Cost-Complexity Pruning

### 6.1.1 Method description

[TODO — Describe cost-complexity pruning: `cost_complexity_pruning_path` to get candidate `ccp_alpha` values, alpha selection by cross-validation on TRAIN ONLY (never on test), plus the required grid over `max_depth`  $\in \{5, 8, 10, 15\}$  and `min_samples_leaf`  $\in \{1, 5, 10, 20\}$ . State explicitly that test accuracy was never used to pick hyperparameters – this is the one thing an instructor is guaranteed to check.]

### 6.1.2 Modified tree / model setting

[TODO — State the final chosen configuration (`ccp_alpha`, `max_depth`, `min_samples_leaf`, `random_state=42`) and the CV accuracy that selected it.]

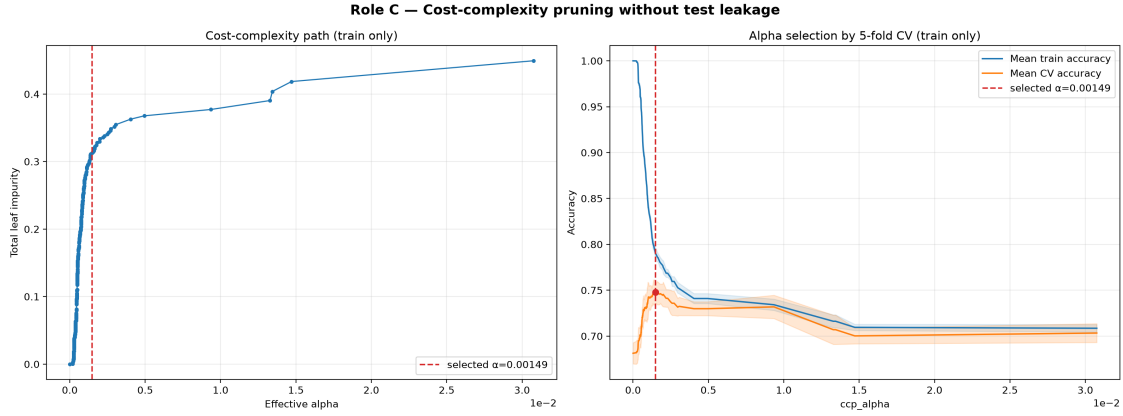


Figure 9: [TODO — Caption: left panel = leaf impurity vs. alpha; right panel = train/CV accuracy with the selected alpha marked. No test curve, by design.]

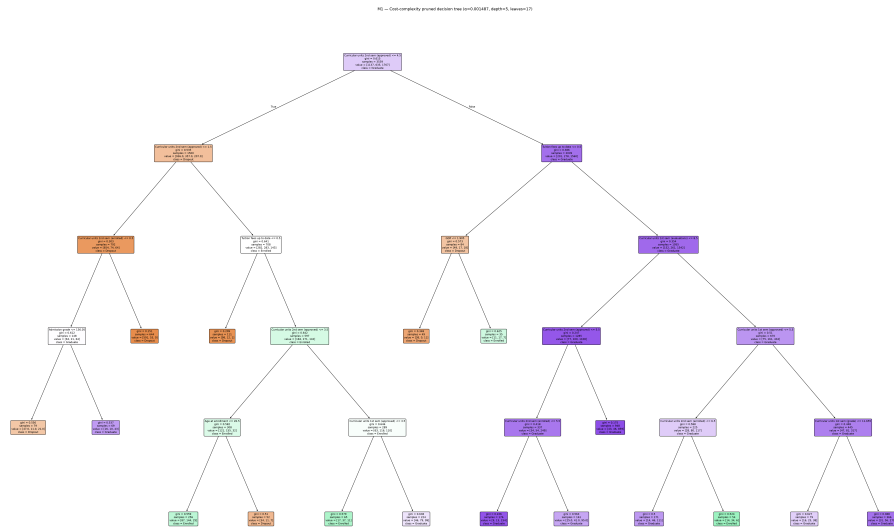


Figure 10: [TODO — Caption: the pruned M1 tree (depth 5, 17 leaves) – small enough to read and use for rule extraction, unlike M0.]

### 6.1.3 Accuracy and error rate

[TODO — Report M1's test\_acc / error\_rate from outputs/results.csv and compare against M0 (test\_acc 0.6689, error\_rate 0.3311). Include the confusion matrix.]

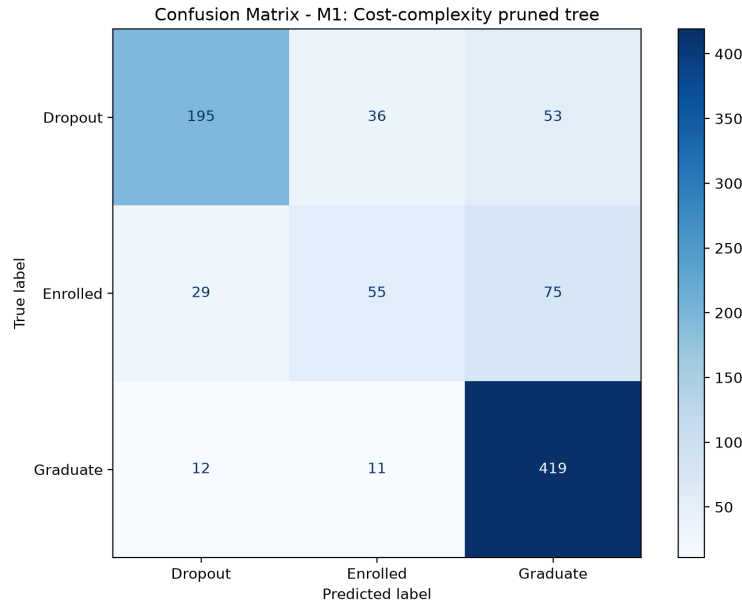


Figure 11: [TODO — Caption: confusion matrix on the same 885-sample held-out test set as M0.]

#### 6.1.4 Why this improves the result

[TODO — Explain why accuracy went up while the tree got dramatically smaller (fewer leaves = less noise memorized, not less signal). Also discuss the one trade-off: which per-class recall got worse, and why (accuracy was the pre-declared CV selection criterion, so the config was free to favor the majority class if that raised the total correct count).]

## 6.2 f.2 Improvement Method 2: Class Imbalance

### 6.2.1 Method description

[TODO — Describe both configurations: M2a = `class_weight='balanced'`; M2b = SMOTE (Chawla et al., 2002; imbalanced-learn developers, 2026), fit *only* on the training split obtained from `src.data.get_train_test()`, never before the split.]

## 6.2.2 Modified tree / model setting

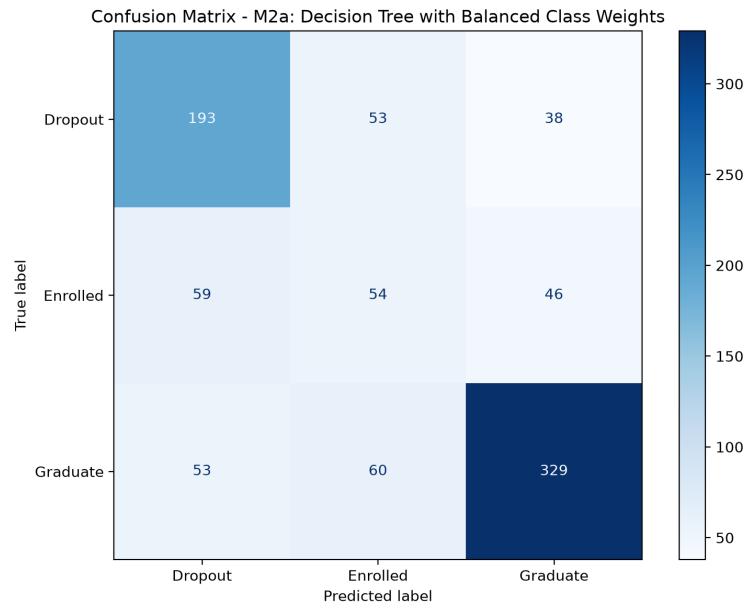


Figure 12: [TODO — Caption: M2a confusion matrix.]

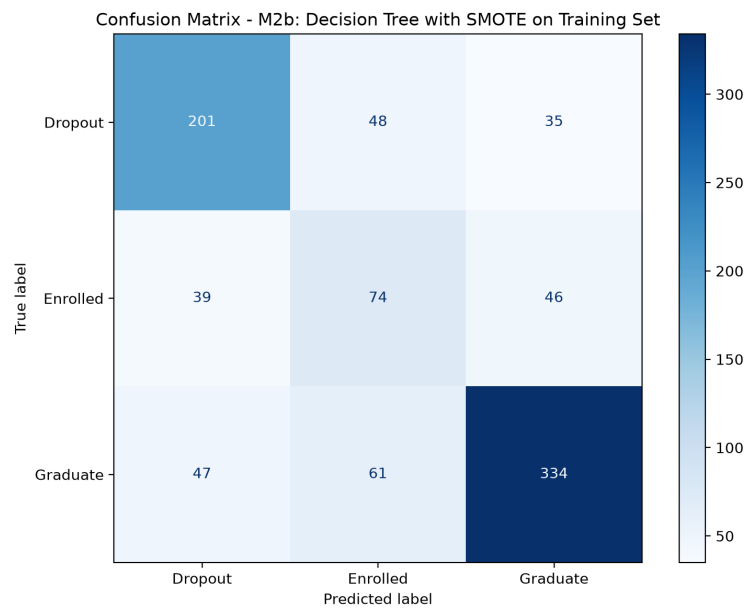


Figure 13: [TODO — Caption: M2b confusion matrix.]

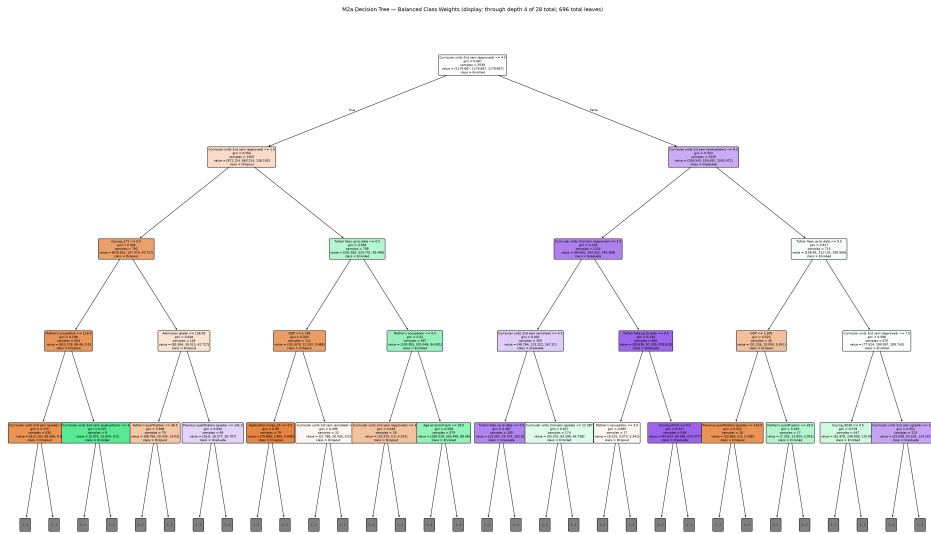


Figure 14: [TODO — Caption: M2a tree, truncated view (state real depth/leaves and note this view's depth limit).]

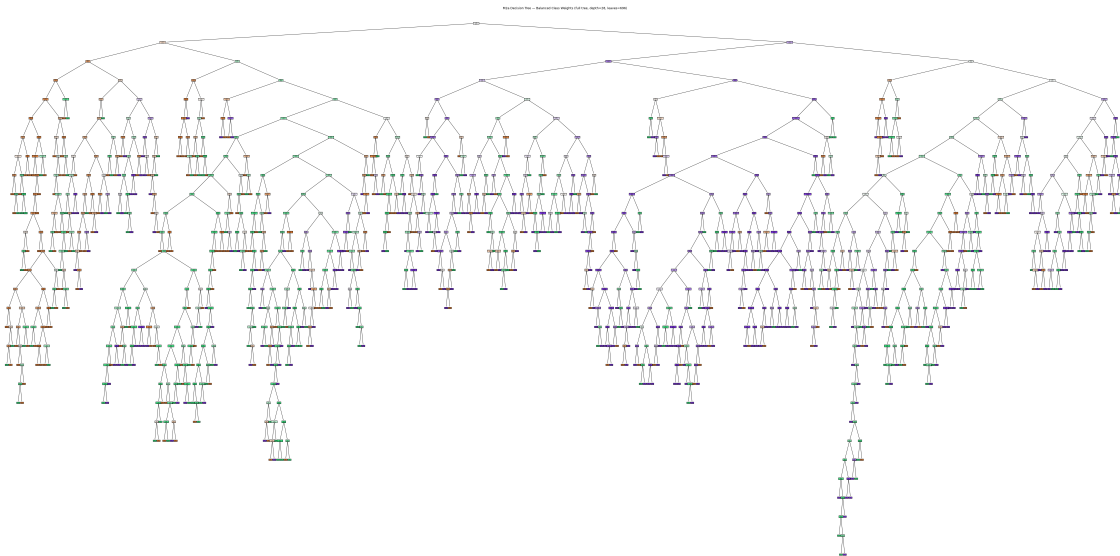


Figure 15: [TODO — Caption: full M2a tree – state it is roughly as complex as M0, since class\_weight changes split priorities, not tree size.]

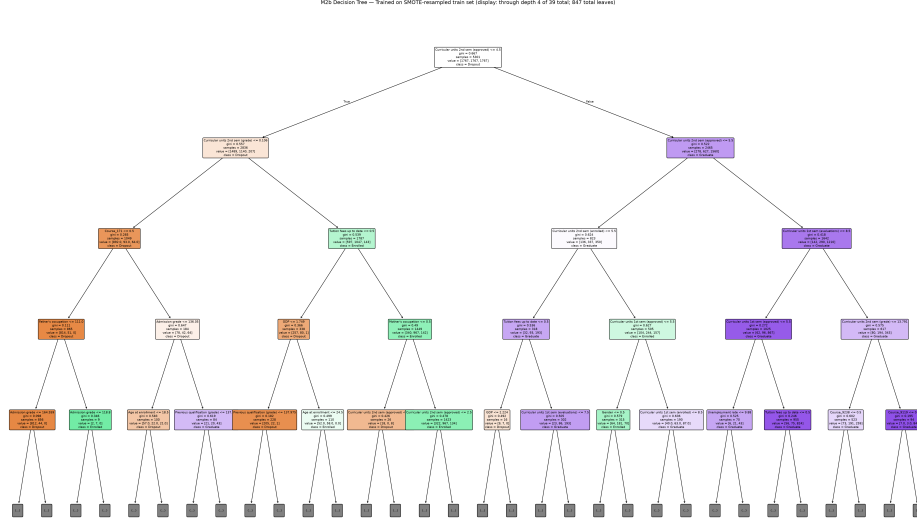


Figure 16: [TODO — Caption: M2b tree, truncated view – note it was trained on the SMOTE-resampled train set but evaluated on the original test set.]

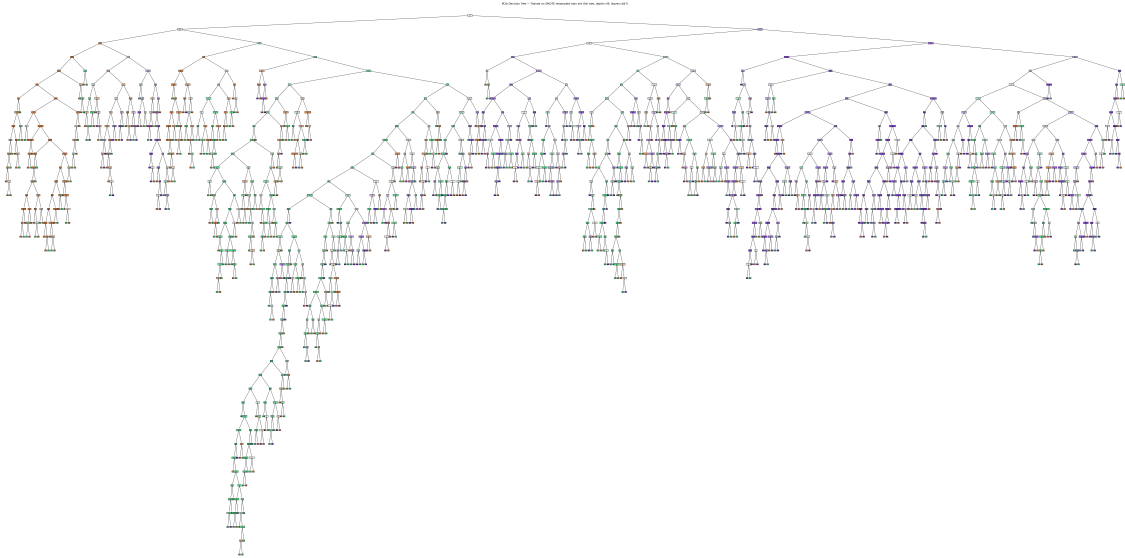


Figure 17: [TODO — Caption: full M2b tree – larger than M0/M2a because SMOTE grows the training set (3,539 → 5,301 rows), giving the tree more rows to split on.]

### 6.2.3 Accuracy and error rate

[TODO — Report M2a and M2b test\_acc / error\_rate against M0 from outputs/results.csv, plus the per-class recall table (Dropout / Enrolled / Graduate) for M0 vs. M2a vs. M2b – this table is the actual point of the experiment, not overall accuracy.]

### 6.2.4 Why this does (or does not) improve the result

[TODO — State plainly that M2a does *not* help on this split (both accuracy and Enrolled recall drop) – do not hide this. Explain M2b’s improvement mechanism (synthetic minority-class samples give the tree more opportunities to find a separating split), and be explicit that recall Enrolled/Dropout going up while overall accuracy is a secondary concern is the correct way to judge success here, per AGENT.md’s own guidance that overall accuracy is not the target metric for this improvement.]

[TODO — Optional but already written and verified in the source draft: the SMOTE-on-one-hot-columns limitation (vanilla SMOTE interpolates one-hot dummy vectors, and imbalanced-learn’s dtype restoration then truncates – not rounds – the fractional interpolated values back to int, producing invalid all-zero one-hot rows for a measurable fraction of synthetic samples, worst for Course at 83.0%). Worth including as a stated limitation even though it does not invalidate the recall/precision gains, which were measured on the real held-out test set.]

### 6.3 f.3 Improvement Method 3: Feature Selection for Early Warning

### 6.3.1 Method description

**[TODO — Describe the method: drop exactly the 12 semester-outcome columns (all six Curricular units 1st sem (...) and six Curricular units 2nd sem (...) columns), keep the remaining 24 original features including the 3 macroeconomic variables (known at enrollment time), and train an unconstrained DecisionTreeClassifier(random\_state=42) exactly like M0 so the only experimental difference is feature availability.]**

### 6.3.2 Modified tree / model setting

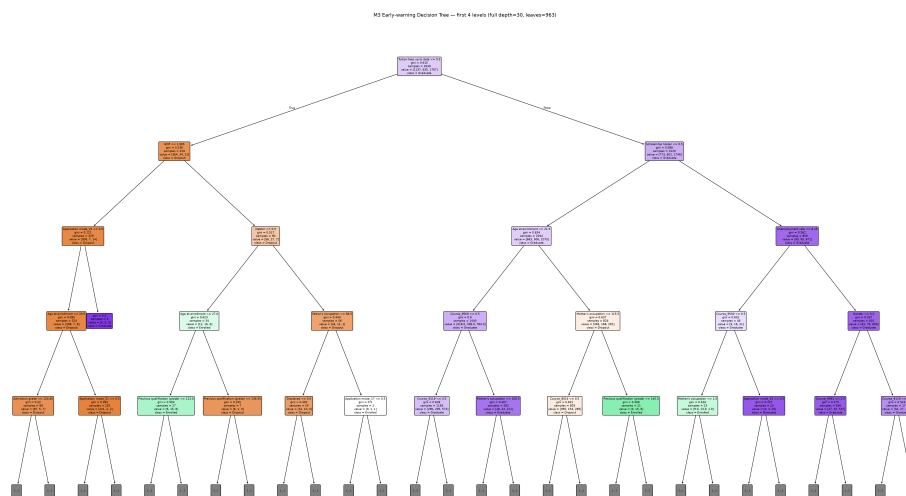


Figure 18: **[TODO — Caption: M3 tree (state depth/leaves from outputs/results.csv).]**

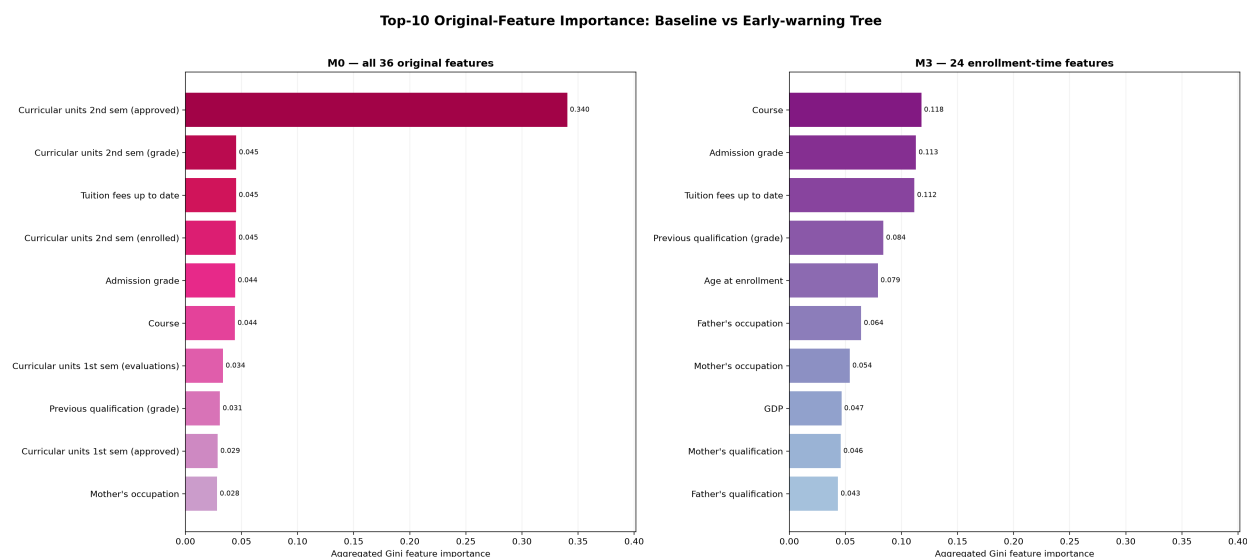


Figure 19: [TODO — Caption: top-10 Gini feature importance, M0 (all 36 original features) vs. M3 (24 retained features) side by side, one-hot dummy importances summed back to their original UCI feature name for a fair comparison.]

### 6.3.3 Accuracy and error rate

[TODO — Report M3's test\_acc / error\_rate / F1 macro against M0 from outputs/results.csv, and the confusion matrix below.]

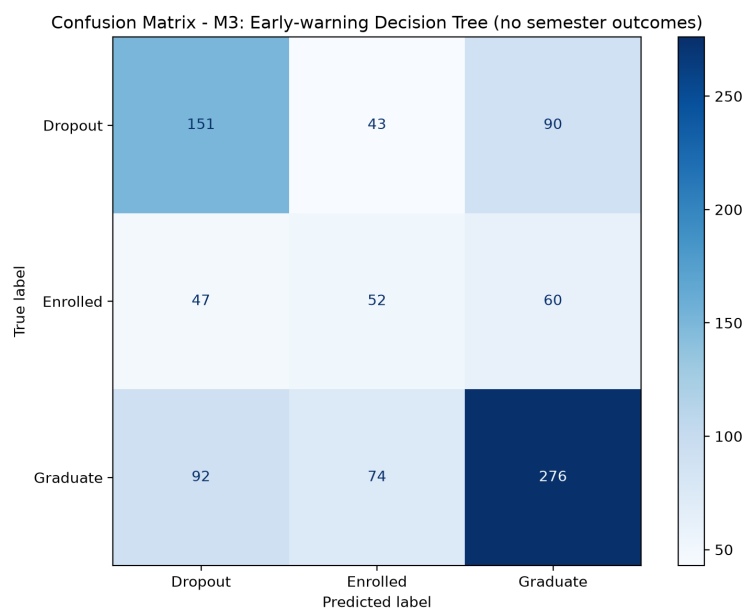


Figure 20: [TODO — Caption: M3 confusion matrix on the same 885-sample held-out test set as every other model.]

### 6.3.4 Why this does not improve accuracy — and why that is expected

[**TODO** — Explain plainly that accuracy drops substantially (this is the intended finding, not a failure): the 12 dropped columns include several of M0’s strongest actual Gini-importance features (see `outputs/E_feature_importance_comparison.csv` – 2nd sem approved is M0’s single most important feature by a wide margin), all of which are only known after a semester ends. State the trade-off explicitly: M0 is more accurate but only actionable after the fact; M3 is less accurate but is the only model of the four that is actually usable at enrollment time for early, preventive intervention. Cite (Martins et al., 2021) here if used as background for the early-prediction framing.]

## 7 g. Comparison of Results

### 7.1 Overall comparison table

Table 5: All 5 models on the shared 885-sample held-out test set

| Model              | Test acc      | Error         | F1 macro      | ROC-AUC       | Rec. Dropout  | Rec. Enrolled | Rec. Graduate | Depth    | Leaves    |
|--------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|----------|-----------|
| M0 (baseline)      | 0.6689        | 0.3311        | 0.6083        | 0.7195        | 0.6796        | 0.3836        | 0.7647        | 27       | 634       |
| M1 (pruning)       | <b>0.7559</b> | <b>0.2441</b> | <b>0.6729</b> | <b>0.8477</b> | 0.6866        | 0.3459        | <b>0.9480</b> | <b>5</b> | <b>17</b> |
| M2a (class_weight) | 0.6508        | 0.3492        | 0.5854        | 0.7053        | 0.6796        | 0.3396        | 0.7443        | 28       | 696       |
| M2b (SMOTE)        | 0.6881        | 0.3119        | 0.6387        | 0.7421        | <b>0.7077</b> | <b>0.4654</b> | 0.7557        | 39       | 847       |
| M3 (early warning) | 0.5412        | 0.4588        | 0.4931        | 0.6254        | 0.5317        | 0.3270        | 0.6244        | 30       | 963       |

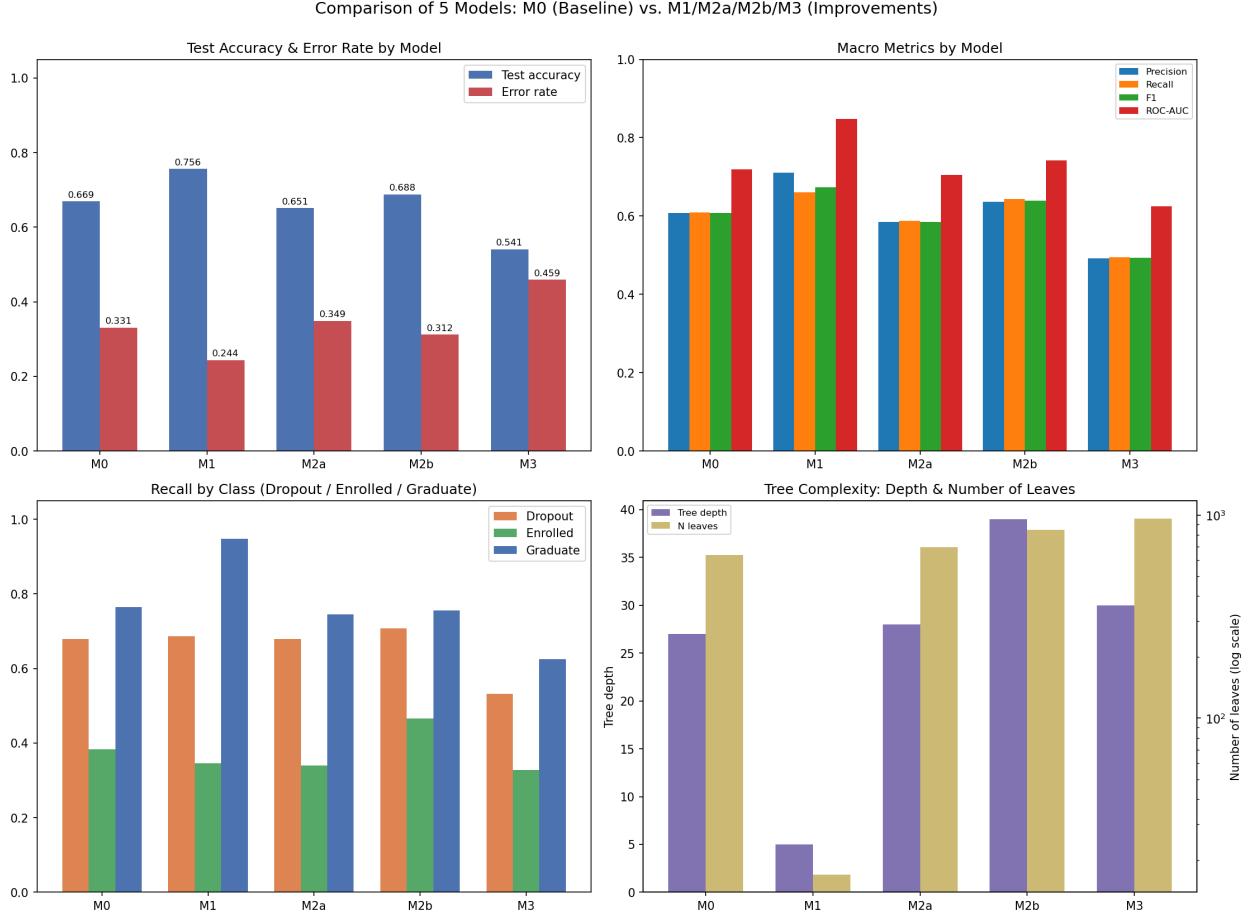


Figure 21: Four-panel comparison of M0-M1-M2a-M2b-M3 on the same held-out test set (885 samples).

Reading the four panels: top-left (test accuracy and error rate) shows M1 highest and M3 lowest; top-right (the four macro metrics – Precision/Recall/F1/ROC-AUC) shows M1 leading on all four; bottom-left (per-class recall) shows M2b dominating on Dropout and Enrolled while M1 dominates on Graduate; bottom-right (tree complexity, leaf count on a log scale) shows M1 far more compact than the other four models.

## 7.2 Comparing each improvement against the baseline

**M1 (Pruning) improves almost every aggregate metric.** Test accuracy rises from 0.6689 to 0.7559 (+**8.70 percentage points**), error rate drops from 33.11% to 24.41%, F1 macro rises 6.47 points, and ROC-AUC macro rises 12.82 points. At the same time, complexity drops sharply: depth 27→5, leaves 634→17 (**about 37× fewer**). This is a rare outcome in practice: the improvement increases both generalization performance and interpretability at once, with no trade-off. The one cost is a slight drop in Enrolled recall (0.3836→0.3459, −3.77 points), because accuracy on CV was the pre-declared criterion for selecting `ccp_alpha`/`max_depth`/`min_samples_leaf`, so the winning configuration was free to favor the majority class (Graduate) whenever that raised the total correct-prediction count.

**M2 (Class Imbalance) gives two opposite results**, consistent with the experimental nature of this improvement. M2a (`class_weight='balanced'`) **does not improve** M0 on this test set:

test accuracy drops 1.81 points, and even Enrolled recall – the metric this technique targets – also drops (0.3836→0.3396). In contrast, M2b (SMOTE on train only) improves both overall accuracy (+1.92 points) and the recall of the two hardest classes: Dropout recall +2.82 points, Enrolled recall +**8.18 percentage points** (61/159 → 74/159 Enrolled students correctly identified), at the cost of a slight 0.90-point drop in Graduate recall. M2b also raises Enrolled precision relative to M0 (0.3567→0.4044), meaning the model is not simply “guessing Enrolled more often” to buy recall – this is a genuine improvement, not an artifact of biased prediction.

**M3 (Early Warning) drops performance substantially, and this is the expected result, not a failure.** Test accuracy falls from 0.6689 to 0.5412 (–12.77 **percentage points**), F1 macro falls 11.52 points. The direct cause, measured from M0’s own tree (not inferred from the original paper): by M0’s real Gini feature importance (computed by Role E), 3 of the top 5 features are **2nd sem approved** (rank 1, importance 0.340 – over a third of the whole tree’s total importance), **2nd sem grade** (rank 2, 0.045), and **2nd sem enrolled** (rank 4, 0.045) – all three fall inside the 12 columns dropped in M3 because they are only known *after* the second semester ends. The other two features in M0’s real top-5 – **Tuition fees up to date** (rank 3) and **Admission grade** (rank 5) – are retained in M3 since they are known at enrollment, but that is not enough to make up for losing **2nd sem approved**, which alone accounts for more importance than the next four features combined. M3 must predict using only information available at enrollment time (program of study, admission grade, tuition status, the 3 macroeconomic variables, ...).

*Note: the top-5 feature table reported by Realinho et al. (2022) (which lists 1st sem approved at rank 2) reflects permutation importance from Random Forest/XGBoost/LightGBM/CatBoost in the original paper – not this project’s own M0 tree’s Gini importance. The two tables ranking differently is expected since they come from different algorithms: 1st sem approved ranks only 9th in this project’s own M0 (importance 0.029), not in the real top-5 of the baseline model. The paragraph above uses only self-measured numbers, not values borrowed from Section A5.*

### 7.3 Which model is “best”?

There is no single answer – it depends on the criterion:

Table 6: Best model by criterion

| Criterion                                        | Best model | Why                                                                 |
|--------------------------------------------------|------------|---------------------------------------------------------------------|
| Overall accuracy / F1 / ROC-AUC                  | <b>M1</b>  | Highest on nearly every aggregate metric, and the most compact tree |
| Detecting Dropout students                       | <b>M2b</b> | Highest Dropout recall (0.7077)                                     |
| Detecting Enrolled students (hardest class)      | <b>M2b</b> | Highest Enrolled recall (0.4654), ~1.21× M0                         |
| Interpretability / presentation                  | <b>M1</b>  | Only 17 leaves, fully readable, vs. 634–963 for the other models    |
| Early deployment (before semester results exist) | <b>M3</b>  | The only model that needs no semester-1/2 data                      |

**On accuracy alone, M1 wins.** But the main lesson of comparing all four models is that “best” depends on the application goal. M1 is best for an accurate, interpretable classification system used once semester data already exists. M2b is best if the goal is to miss as few at-risk

students as possible (Dropout/Enrolled), accepting a small overall-accuracy trade-off. M3 is the only feasible choice if the goal is **early intervention right at enrollment** – none of the other three models can run at that point in time, no matter how much higher their accuracy is.

## 8 h. Conclusion

### 8.1 What the group learned

The project directly demonstrates the two theoretical points raised in the Introduction: a decision tree is both powerful (interpretable, handles mixed data natively) and prone to overfitting if its complexity is not controlled. The unconstrained M0 model reaches perfect training accuracy (1.000) while test accuracy is only 0.6689 – a generalization gap of 0.3311 that is direct, not inferred, evidence of the overfitting phenomenon described in the theory (Breiman et al., 1984; Mitchell, 1997).

The three improvement directions teach three distinct lessons, with no overlap between them:

1. **Controlling complexity (pruning) is almost always beneficial** when the original model is badly overfit – not a trade-off, but a simultaneous improvement in both accuracy and interpretability.
2. **Class-imbalance techniques do not automatically help** – effectiveness depends on the specific mechanism (M2a reweights the Gini computation without adding data, while M2b adds synthetic samples in the minority class’s feature region), and must be judged by per-class recall, not overall accuracy alone – a model can be “worse” on accuracy yet “better” on the real application goal.
3. **The most accurate model is not always the most useful one.** M3 shows that this dataset’s strongest predictive features (semester outcomes) are only available once it is already too late for preventive intervention – so M3’s lower accuracy is a reasonable price for early-warning capability.

### 8.2 Key findings from the experiments

- M0’s overfitting: a 0.3311 generalization gap (train 1.000 vs. test 0.6689), depth 27, 634 leaves – the tree “memorizes” rather than generalizes.
- Pruning shrinks the leaf count by  $\sim 37\times$  (634 $\rightarrow$ 17) while still raising test accuracy by 8.70 percentage points – evidence that most of M0’s 634 leaves described noise, not signal.
- `class_weight='balanced'` (M2a) does not improve minority-class recall on this data; SMOTE-on-train (M2b) raises Enrolled recall by 8.18 points and Dropout recall by 2.82 points, while also raising precision – a genuine improvement, not an artifact.
- Dropping the 12 semester-outcome columns (M3) costs 12.77 accuracy points – a direct measurement of the “price” of wanting an early warning instead of a later but more accurate one.
- All 5 models share one fixed train/test split (`random_state=42`, stratified), so every difference reported above reflects a genuine difference between methods, not noise from different data splits.

### 8.3 Effectiveness of decision trees for this problem

A decision tree suits this dataset for the reasons argued in the Dataset Description section: features have clear semantics, so every decision rule is interpretable – for example, the Dropout rule extracted from the M0 tree in Section 5 is a chain of 9 conditions that together identify a leaf of  $n = 189$  training samples at 100% purity; the single most readable condition in that chain is passing at most 1.5 second-semester course units (`2nd sem approved`  $\leq 1.5$ ), but that threshold alone is not sufficient on its own – it is one term in the full conjunction. The parallel Graduate leaf ( $n = 176$ , 100% purity) is a 16-condition chain, the two most readable conditions being passing more than 5.5 second-semester course units *and* Tuition fees up to date  $> 0.5$  – the latter consistent with the EDA observation in Figure 2 that paying tuition on time correlates strongly with graduating. Long condition chains like these are themselves a sign of the overfitting discussed above: a rule that needs 9–16 ANDed conditions to stay 100% pure is fitting narrow pockets of the training set, not a simple general pattern – one of the concrete costs of leaving M0 unpruned. The tree also needs no feature scaling and handles categorical and numerical features natively. But the project also exposes the inherent limits of a single, unconstrained tree: it overfits easily, and no single model configuration is simultaneously the most accurate, the fairest across classes, and the earliest-usable – these three goals require three different configurations of the same algorithm. For an application with multiple distinct goals, like dropout prediction, a more realistic conclusion than “one best model” is: **a decision tree is a flexible enough tool to be optimized for each goal separately** (accuracy, class fairness, or timeliness), provided the user chooses the right configuration and understands the trade-off being accepted – exactly the spirit the assignment brief asks for: not just running a model, but explaining why each choice does or does not help.

## References

- Leo Breiman, Jerome H. Friedman, Richard A. Olshen, and Charles J. Stone. *Classification and Regression Trees*. Wadsworth, 1984.
- Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, 2002.
- imbalanced-learn developers. imblearn.over\_sampling.SMOTE. imbalanced-learn 0.14.2 documentation, 2026. URL [https://imbalanced-learn.org/stable/references/generated/imblearn.over\\_sampling.SMOTE.html](https://imbalanced-learn.org/stable/references/generated/imblearn.over_sampling.SMOTE.html). Accessed 2026.
- Mónica V. Martins, Daniel Tolentino, Jorge Machado, Luís Baptista, and Valentim Realinho. Early prediction of student’s performance in higher education: A case study. In *Trends and Applications in Information Systems and Technologies (WorldCIST 2021)*, volume 1365 of *Advances in Intelligent Systems and Computing*, pages 166–175, 2021. Background reading for M3 early-warning rationale; not yet cited in the current f.3 draft – add [Martins et al. \(2021\)](#) in sections/f3\_features.tex if used.
- Tom M. Mitchell. *Machine Learning*. McGraw-Hill, 1997.
- Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas,

- Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12: 2825–2830, 2011.
- J. Ross Quinlan. Induction of decision trees. *Machine Learning*, 1(1):81–106, 1986.
- Valentim Realinho, Mónica Vieira Martins, Jorge Machado, and Luís Baptista. Predict students’ dropout and academic success. UCI Machine Learning Repository, 2021. Dataset, id=697.
- Valentim Realinho, Jorge Machado, Luís Baptista, and Mónica V. Martins. Predicting student dropout and academic success. *Data*, 7(11):146, 2022. doi: 10.3390/data7110146.